Noah Adasi

1 University Avenue, Berekuso, Ghana, PMB CT 3 • noahadasi6@gmail.com||noah.adasi@alumni.ashesi.edu.gh
(+233) 248726507 • https://adasinoah.github.io/ • LinkedIn • YouTube

Education

Ashesi University Berekuso, Ghana
BSc. Electrical & Electronic Engineering (EQF Level 6) 07/09/2020 – 16/08/2024
Thesis: Machine Learning-Based Electronic Scarecrows for Protecting Agricultural Yields from Predator Damage
Core Background: Power Engineering, Control Systems, Electrical Machines, Communication Systems & Advanced Communication Systems, Signal Processing, Digital Signal Processing, Power Electronics, Differential Equations & Numerical Methods, Circuit z& Electronics (other courses taken)

Research Experience

Ashesi University Berekuso, Ghana
Teaching & Research Assistant (Faculty Intern) 02/09/2024 – Present (2 years)

- Serve as Teaching Assistant for graduate Mechatronics (MSc.) – Signals and Systems and undergraduate courses: Communication Systems, Advanced Communication Systems, Digital Signal Processing, Signals and Systems, and Engineering Calculus
- Lead study and laboratory sessions, grade assignments, conduct office hours, and support faculty research

African Leadership Academy South Africa
AI for Health Research Fellow 01/10/2024 – 30/03/2025

- Conduct research on accessing water quality for treated drinking water in galamsey (illegal mining) affected areas in Ghana
- Collaborate with experienced healthcare research professionals for coaching and mentorship
- Participate in structured learning modules to enhance knowledge in critical areas of medical research in Africa

Technical & Research Projects

- **Machine Learning-Based Electronic Scarecrow for Protecting Agricultural Yields** (Capstone Project, 01/2024 – 08/2024): Trained a YOLOv8 model using customized datasets to analyze real-time camera feeds for detecting and classifying bird threats; designed and integrated responsive bird-scaring mechanisms; created a prototype combining traditional deterrent methods with innovative technologies
- **FPGA-Based FFT Hardware Accelerator for Blur Detection and Image Deconvolution in UAV Surveillance** (Co-author): Accepted for presentation at the 6th Interdisciplinary Conference on Electrics and Computer (INTCEC 2026), Chicago, USA, 24–25 September 2026
- **Cooperative Spectrum Sensing Using Cognitive Radio** (05/2024 – 07/2024): Implemented Cooperative Spectrum Sensing algorithms in MATLAB and Python to enhance channel detection accuracy; conducted simulations to evaluate cognitive radio effectiveness in dynamic spectrum management
- **Digital Signal Processing with Pulse Code Modulation (PCM)** (11/2023 – 12/2023): Analyzed PCM stages (sampling, quantization, encoding) to quantify effects on signal fidelity; utilized MATLAB and Simulink to simulate PCM processes and recommend optimized bit depths
- **Design and Implementation of a Three-Phase Automatic Changeover Switch**: Designed and implemented a system using contactors, relays, timers, and switches for automatic transfer between ECG and generator power; published in the *Ashesi Research Seed Journal*
- **CNN Machine Learning Model for Option Trading** (03/2024 – 04/2024): Developed a CNN in TensorFlow to predict option trading outcomes based on financial ratios; optimized model architecture through iterative adjustments to layers and nodes

- **Optimization Using Gradient** (11/2021 – 12/2021): Investigated the q-gradient method using Jackson's derivative to enhance classical gradient techniques for unconstrained global optimization; conducted comparative analyses against Genetic Algorithms
- **Design of a Rectifier & DC-DC Buck Converter**: Designed and tested circuits to convert 230Vrms AC to stable 48V DC with high efficiency (unregulated supply reached 91%)
- **Design and Control of Micro Power Grids**: Explored hierarchical control architectures, renewable energy integration, and challenges in microgrid operation for improved resilience and sustainability
- **Optical Heartbeat Monitor Instrument**: Designed a system using IR LED, photodiode, filters, amplifiers, ADC, and Arduino to measure heart rates from 68 to 180 bpm
- **7-Segment Display Based Digital Clock** and **Digital LED Clock**: Designed and implemented fully functional digital clock circuits

Publications

- Afriyie, A., Aho, E., Kuuchi, D., Osei, W., & Adasi, N. Design and Implementation of a Three-Phase Automatic Changeover Switch Using CAD Electrical Simulation. *Ashesi Research Seed Journal*.

Conferences & Presentations

6th Interdisciplinary Conference on Electrics and Computer (INTCEC 2026) Chicago, USA
Accepted Co-Author & Presenter 24–25 September 2026

- "FPGA-Based FFT Hardware Accelerator for Blur Detection and Image Deconvolution in UAV Surveillance"

British Conference of Undergraduate Research (BCUR 2026) Glasgow, United Kingdom
Oral Presenter 01–02 April 2026

- Delivered oral presentations on "Signal Processing for AI Privacy in Machine Learning: Input-First Quantization-Based Training Technique" and "Machine Learning-Based Electronic Scarecrows for Protecting Agricultural Yields from Predator Damage"

Council on Undergraduate Research (CUR) Richmond, Virginia, United States
Accepted Presenter 13–15 April 2026

- Selected to present research on "Assessing the Quality of Treated Drinking Water in Galamsey-Affected Areas in Ghana" and "The Role of Climate Finance in Developing Countries: A Comprehensive Analysis of Multilateral and Blended Climate Finance for Adaptation in Africa"

Professional Experience

EducationUSA – ACE Consult Limited Accra, Ghana
GRE, GMAT, and SAT Mathematics Lecturer 01/10/2024 – Present

- Deliver instruction in GRE, GMAT, and SAT Math to improve students' problem-solving and test performance
- Develop tailored lesson plans and materials to cover each exam's specific mathematics requirements
- Monitor students' progress and adapt teaching methods to support goal achievement on standardized tests

Ashesi University Berekuso, Ghana
Campus Ambassador 01/10/2021 – Present

- Assist the Admissions Office with campus tours and provide prospective students and visitors with insights into Ashesi's programs and culture
- Promote Ashesi University's mission and values through outreach and engagement activities
- Support admissions events and presentations; collaborate with faculty and staff to offer personalized campus visits

Jospong Groups of Companies – Sewerage Systems Ghana Ltd Accra, Ghana
Electrical & Electronic Engineering Intern 01/06/2023 – 30/08/2023

- Installed and configured attendance tracking systems, including electronic locks for workforce monitoring
- Performed forward and reverse motor control wiring to support operational automation
- Assisted in troubleshooting and resolving electrical issues on-site to ensure system functionality

Musizi University Kampala, Uganda
Educational Research Intern 03/06/2022 – 30/08/2022

- Collected and analyzed data on scholarship-providing NGOs and identified leading private and government schools in Uganda
- Designed an Entrepreneurship & Programming curriculum for the University's Innovation Camp
- Developed student handbooks and outlined a constitution to support governance and organizational structure

Leadership & Activities

- Founder & President, Galaxy Space Campus Network, Ashesi University (promotes space education and representation of Ghanaian and African students)
- Founder, NPath Foundation (focused on sustainable community development and education support for underprivileged students)
- Chairperson, Electoral & Documentation Committee, MasterCard Foundation – Ashesi University (2021–2022)
- Chairperson, Feedback Committee, MasterCard Foundation – Ashesi University
- Board Member, Ashesi University Student Council Academic Committee (2021–2023)
- Campus Ambassador, Ashesi University (2021–Present)
- Peer Tutor (Engineering Calculus and Applied Calculus)
- Event Management Board Member, TEDxAshesiUniversity
- Class Representative, Corporate Ethics in Africa Course

Awards & Certifications

- MasterCard Foundation Scholars Program Scholarship, Ashesi University (2020–2024)
- Global Mentorship Initiative (GMI) Program (2023) – Mentored by a senior industry professional from Brazil
- Certificate of Honor – Electric Industry Operations and Markets, Duke University (Coursera)
- Certificate of Honor – Introduction to Electronics, Georgia Institute of Technology (Coursera)
- Ghana National Science and Math Quiz – National Level Contestant Honors, Sunyani Senior High School (2019)

Technical Skills & Research Interests

Technical Skills:
PowerWorld, Proteus, EasyEDA (PCB Design); Python, PyTorch, TensorFlow, MATLAB, C#, Scikit-learn; Deep Learning, Reinforcement Learning, LLM Agent Architectures, Probabilistic Modeling; Digital Signal Processing, Cooperative Spectrum Sensing; Git/GitHub, Docker, Google Cloud Platform (GCP); FPGA-based accelerators

Research Interests:
Optimization and control of distributed energy resources, demand response, and resilient power systems; signal processing techniques for privacy-preserving and robust machine learning (including reinforcement learning); applied computer vision for agriculture and surveillance; microgrid control; 5G/6G communications; ethical and reliable AI systems

Professional Affiliations

IEEE Student Member (No. 100767403)
Member, IEEE Young Professionals & Signal Processing Society